

Immunodeficiency 97 with Autoinflammation (IMD97): A Comprehensive Disease Characteristics Report

Disease: Immunodeficiency 97 with autoinflammation (IMD97) **Gene:** *PIK3CG* (biallelic loss-of-function) — encodes the p110 γ catalytic subunit of PI3K γ **Inheritance:** Autosomal recessive **MONDO:** MONDO:0030717 | **OMIM (phenotype):** 619802 | **OMIM (gene):** 601232 **Explicit scope note:** This report concerns the *PIK3CG*-related inborn error of immunity. It is **NOT** TBK1 deficiency and is **NOT** *PIK3CD*-related Activated PI3K δ Syndrome (APDS).

Summary

Immunodeficiency 97 with autoinflammation (IMD97) is an ultra-rare, autosomal-recessive inborn error of immunity (IEI) caused by biallelic loss-of-function (LOF) variants in *PIK3CG*, the gene encoding p110 γ — the sole class IB phosphoinositide 3-kinase (PI3K) catalytic subunit. p110 γ is expressed predominantly in leukocytes (myeloid cells, T cells, and B cells) and is uniquely activated downstream of G-protein-coupled receptors (GPCRs, via G $\beta\gamma$) and Ras to generate the second-messenger lipid PIP3. The disease was first defined in a single human patient in 2019 ([PMID: 31554793](#)) and independently confirmed in a second report in 2020 ([PMID: 33054089](#)). It was formally admitted as a distinct IEI entity by the International Union of Immunological Societies (IUIS) Expert Committee.

The hallmark of IMD97 is a **paradoxical dual phenotype**: a **humoral immunodeficiency** (impaired B-cell/antibody-secreting-cell differentiation, hypogammaglobulinemia, poor vaccine responses, recurrent sinopulmonary infections, cytopenias) coexisting with **myeloid-driven autoinflammation and tissue immunopathology** (T-lymphocytic pneumonitis, colitis). Mechanistically, loss of p110 γ removes a cell-intrinsic signal required for B cells to differentiate into antibody-secreting cells ([PMID: 38961274](#)), while simultaneously de-repressing a GSK3 α/β -dependent program in macrophages and monocytes that drives

excessive IL-12/IL-23 production upon Toll-like receptor (TLR) stimulation ([PMID: 31554793](#)). The autoinflammatory tissue disease is microbiota-dependent: *Pik3cg*-deficient mice only recapitulate the human phenotype after exposure to natural microbiota (co-housing with pet-store mice).

This report synthesizes eight confirmed findings and 54 reviewed papers across all 15 requested disease-characteristic domains. Because IMD97 is described in only a handful of patients worldwide, several clinical domains (formal epidemiology, prognosis statistics, trial-based treatment data) rest on case reports and mechanistic extrapolation rather than cohort evidence — these gaps are flagged explicitly throughout. The genetic reality of the entity is nonetheless firmly established: ClinVar curates specific pathogenic missense, nonsense, and frameshift *PIK3CG* LOF alleles to IMD97, and gnomAD constraint metrics (pLI $\approx$ 0) confirm heterozygous carriers are healthy — consistent with the recessive model.

Key Findings

F001 — IMD97 is caused by biallelic loss-of-function *PIK3CG* variants that abolish the p110 γ catalytic subunit

The first human patient, reported by Takeda et al. (2019, *JCD*), carried **bi-allelic, loss-of-function mutations in *PIK3CG* resulting in absence of the p110 γ catalytic subunit of PI3Ky**. Her clinical history comprised **childhood-onset antibody defects, cytopenias, and T-lymphocytic pneumonitis and colitis** ([PMID: 31554793](#)). A second, independent report by Thian et al. (2020, *JACI*) described **germline biallelic *PIK3CG* mutations in a multifaceted immunodeficiency with immune dysregulation** ([PMID: 33054089](#)), confirming the entity and its autosomal-recessive inheritance.

PIK3CG maps to chromosome **7q22.3** (GRCh38 chr7:106,865,278–106,908,980, + strand); the protein product is UniProt **P48736** (PK3CG_HUMAN); HGNC:8978; Ensembl ENSG00000105851. This finding establishes both the genetic cause (biallelic LOF, loss of p110 γ) and the core clinical triad — humoral immunodeficiency, cytopenias, and organ-specific lymphocytic inflammation.

Evidence source: human clinical (two independent index reports).

F002 — PI3Ky loss dysregulates macrophage/monocyte cytokine output (elevated IL-12/IL-23) via GSK3 α/β , driving autoinflammation

Takeda et al. showed that **PI3Ky-deficient macrophages and monocytes produce elevated inflammatory IL-12 and IL-23 in a GSK3 α/β -dependent manner upon TLR stimulation** (PMID: 31554793). At the cellular level the patient showed reduced memory B cells, memory CD8⁺ T cells, and regulatory T (Treg) cells, with increased CXCR3⁺ tissue-homing CD4 T cells — a signature consistent with skewing toward inflammatory, tissue-infiltrating adaptive responses.

Critically, the mouse model established causality and environmental dependence: **Pik3cg-deficient mice recapitulate major features of human disease after exposure to natural microbiota through co-housing with pet-store mice.** This positions the myeloid IL-12/IL-23 axis as the mechanistic engine of the autoinflammatory arm of IMD97 and identifies GSK3 α/β as the intermediate node de-repressed when p110 γ signaling is lost.

Evidence source: human clinical + model organism (mouse).

F003 — PI3Ky is cell-intrinsically required for B-cell differentiation into antibody-secreting cells, explaining the humoral defect

Lanahan et al. (2024, *Nature Immunology*) demonstrated that **the inborn error of immunity caused by human deficiency in PI3Ky results in broad humoral defects**, and that **PI3Ky functions cell-intrinsically within activated B cells in a kinase-activity-dependent manner to transduce signals required for the transcriptional program supporting differentiation of antibody-secreting cells (ASCs)** (PMID: 38961274). ASC fate coincides with up-regulation of *PIK3CG* expression, and differentiation is impaired when PI3Ky is disrupted in naive B cells, in memory B cells (upon TLR activation), and in human tonsillar organoids.

This finding is pivotal because it resolves the apparent paradox of a "myeloid" kinase causing antibody deficiency: p110 γ has a *direct, kinase-dependent, B-cell-intrinsic* role in the ASC transcriptional program. The humoral immunodeficiency is therefore not merely secondary to inflammation but a primary consequence of losing p110 γ inside the B-cell lineage.

Evidence source: human clinical + in vitro (organoid) + mechanistic.

F004 — Structural/biophysical basis: immunodeficiency mutations at the C-terminal residue R1021 inactivate the enzyme

Rathinaswamy et al. (2021) used hydrogen–deuterium exchange mass spectrometry (HDX-MS) and biophysical assays to **define how immunodeficiency and oncogenic mutations of R1021 in the C-terminus can inactivate or activate enzyme activity** by disrupting regulatory C-terminal dynamics (PMID: 33661099). p110 γ is the class IB PI3K catalytic subunit and a key factor in immune signaling and inflammatory disease. This provides the protein-level, structural mechanism connecting a specific missense change (p.Arg1021Pro, seen in ClinVar; see F008) to loss of catalytic activity — a molecular explanation for how a single-residue change abolishes p110 γ function.

Evidence source: in vitro / biophysical / computational-structural.

F005 — PI3K γ is a myeloid/leukocyte-enriched GPCR effector controlling leukocyte trafficking, Treg homeostasis, and mucosal barrier — loss produces immunodeficiency-with-autoinflammation

Converging mouse genetics establish PI3K γ 's physiological roles: *Pik3cg*-null mice show defective myeloid/monocyte migration and recruitment (PMID: 41794063), altered colitis susceptibility with changes in mucosal IgA/IL-10/occludin and microbiota (PMID: 36031460), reduced neutrophil NET formation and pyroptosis (PMID: 39024551), and roles in Treg/Th17 balance (PMID: 36243853, PMID: 34512641). PI3K γ is "**predominantly expressed in myeloid cells**" and "**plays a pivotal role in mediating inflammatory responses**" (PMID: 41794063); it acts downstream of GPCRs. Takeda et al. summarize the integrated physiology: "**our results emphasize the physiological importance of PI3K γ in restraining inflammation and promoting appropriate adaptive immune responses in both humans and mice**" (PMID: 31554793) — the dual role that produces simultaneous immunodeficiency and immunopathology.

Evidence source: model organism (multiple) + human clinical.

F006 — Class I PI3K framework confirms PI3Ky as the myeloid/leukocyte class IB GPCR-activated isoform generating PIP3

A 2025 review by Shaw, Barlow-Busch & Burke frames the class I PI3K pathway as a master regulator that **"plays essential roles in controlling immune cell function, metabolism, chemotaxis and proliferation,"** noting that activation of class I PI3Ks generates the signalling lipid PIP3 (PMID: 40925445). PI3Ky is uniquely activated by GPCRs (via Gβγ) in leukocytes, supported by Perrotta et al. (2025) showing PI3Ky controls CD8 T-cell trafficking (PMID: 40595568) and structural work on the Ras-PI3Ky interface (PMID: 42148525). This anchors the disease within the well-characterized biochemistry of class I PI3K signaling.

Evidence source: review + in vitro/structural.

F007 — gnomAD constraint (pLI ≈ 0) confirms carrier tolerance, cementing the recessive model

gnomAD constraint metrics for *PIK3CG* (GRCh38) are: **pLI = 3.8×10^{-8} (≈ 0 ; not haploinsufficient)**, observed/expected LOF (oe_lof) = **0.54 (90% CI 0.43–0.67; LOEUF ≈ 0.67)**, missense Z = **1.53**. A pLI near zero means the gene tolerates heterozygous LOF, exactly as expected for an autosomal-recessive disorder in which single-allele carriers are clinically healthy and disease requires biallelic loss. Verified identifiers confirmed in this analysis: HGNC:8978, OMIM gene 601232, UniProt P48736, Ensembl ENSG00000105851, locus chr7:106,865,278–106,908,980 (+ strand), cytoband 7q22.3.

Evidence source: population genomics (gnomAD).

F008 — ClinVar lists specific pathogenic *PIK3CG* variants curated to IMD97, spanning missense, nonsense, and frameshift LOF

ClinVar (reference transcript NM_001282426.2) annotates the following variants to **"Immunodeficiency 97 with autoinflammation"**:

cDNA	Protein	Type	Classification
c.145C>A	p.Arg49Ser	Missense	Pathogenic
c.3254A>G	p.Asn1085Ser	Missense	Pathogenic
c.3062G>C	p.Arg1021Pro	Missense	Pathogenic
c.2545C>T	p.Arg849Ter	Nonsense	Conflicting
c.[391_397del;456_468del]	—	Frameshift/LOF (in-cis)	Pathogenic
c.211A>T	p.Lys71Ter	Nonsense	Likely Pathogenic

The **p.Arg1021Pro** variant maps to the C-terminal R1021 residue shown by Rathinaswamy et al. (PMID: 33661099) to inactivate the kinase, directly linking a curated clinical allele to an experimentally validated inactivation mechanism.

Evidence source: clinical variant database (ClinVar) + biophysical corroboration.

Section-by-Section Disease Characteristics

1. Disease Information

Overview. IMD97 is an autosomal-recessive inborn error of immunity combining a primary antibody deficiency (humoral immunodeficiency) with immune dysregulation and autoinflammation. It arises from complete or near-complete loss of function of PI3Ky, the leukocyte-enriched class IB PI3K. Affected individuals present in childhood with recurrent infections, hypogammaglobulinemia, cytopenias, and organ-specific lymphocytic inflammation (notably pneumonitis and colitis) (PMID: 31554793, PMID: 33054089).

Key identifiers. - **MONDO:** MONDO:0030717 - **OMIM phenotype:** 619802 (Immunodeficiency 97 with autoinflammation) - **OMIM gene:** 601232 (*PIK3CG*) - **HGNC:** 8978 | **UniProt:** P48736 | **Ensembl:** ENSG00000105851 - **Orphanet / ICD-11 / MeSH:** No dedicated Orphanet or specific ICD-11 code was identified for IMD97 as a distinct entity; it falls under broader categories of "combined immunodeficiency with immune dysregulation" / "primary antibody deficiency." (Data not available at the granular level — flagged as a gap.)

Synonyms / alternative names. PI3K γ deficiency; p110 γ deficiency; PIK3CG deficiency; human PI3K γ inborn error of immunity; immunodeficiency-97 (IMD97).

Information source. Disease-level knowledge derives from **individual patient reports** (a small number of index cases) integrated with **aggregated resources** (OMIM, ClinVar, gnomAD) and **mechanistic model-organism/in vitro studies**. This is not an EHR-scale or registry-scale entity.

2. Etiology

Primary cause — genetic. Biallelic (homozygous or compound-heterozygous) LOF variants in *PIK3CG* that abolish or catalytically inactivate p110 γ (F001, F004, F008). The mechanism is loss of function; there is no evidence for gain-of-function or dominant-negative activity in IMD97 (contrast with *PIK3CD* APDS, which is gain-of-function).

Genetic risk factors. The causal variants are the disease determinants; no separate susceptibility loci or GWAS signals apply to this Mendelian condition. Consanguinity increases risk of homozygous LOF (as in recessive IEIs generally).

Environmental factors as disease modifiers. Uniquely, the **microbiota is a required environmental cofactor** for the autoinflammatory tissue disease: *Pik3cg*-deficient mice manifest disease only after exposure to natural microbiota ([PMID: 31554793](#)). Microbial/TLR stimulation is the trigger that unleashes the dysregulated IL-12/IL-23 myeloid response (F002). Infectious exposures also drive the immunodeficiency phenotype (recurrent infections due to antibody failure).

Protective factors. No specific protective alleles or environmental protective factors are established. Heterozygous carriers are unaffected (F007). By extrapolation, a controlled/reduced microbial burden may limit autoinflammatory flares (inferred, not demonstrated in humans).

Gene–environment interaction. The central G×E interaction is **genotype (biallelic *PIK3CG* LOF) × microbiota/TLR ligand exposure → GSK3-dependent IL-12/IL-23 overproduction → tissue immunopathology** (F002). This is one of the clearest documented gene–microbiota interactions among IEIs.

3. Phenotypes

Phenotypes are drawn primarily from the two index reports and mechanistic studies. Frequencies are qualitative given the tiny cohort.

Phenotype	Type	HPO suggestion	Onset	Notes
Recurrent respiratory/sinopulmonary infections	Clinical sign	HP:0002719 (Recurrent infections); HP:0002205 (Recurrent respiratory infections)	Childhood	Consequence of antibody defect
Antibody deficiency / hypogammaglobulinemia	Lab abnormality	HP:0004313 (Decreased circulating antibody level); HP:0002720 (Decreased IgA)	Childhood	Broad humoral defect (F003)
Impaired vaccine/specific antibody responses	Lab abnormality	HP:0005387 (Impaired antibody response)	Childhood	ASC differentiation failure
Cytopenias	Lab abnormality	HP:0001903 (Anemia); HP:0001873 (Thrombocytopenia); HP:0001882 (Leukopenia)	Childhood	Immune dysregulation
T-lymphocytic pneumonitis	Clinical sign / imaging	HP:0006515 (Interstitial pneumonitis)	Childhood	Tissue immunopathology (F001)
Colitis / inflammatory bowel disease	Clinical sign	HP:0002583 (Colitis); HP:0002037 (Inflammatory abnormality of the GI tract)	Childhood	Microbiota-dependent (F002)
Reduced memory B cells	Lab abnormality	HP:0031381 (Decreased circulating memory B cell count)	—	Immunophenotype (F002)

Phenotype	Type	HPO suggestion	Onset	Notes
Reduced regulatory T cells	Lab abnormality	(reduced Treg count)	—	Loss of tolerance
Autoinflammation / immune dysregulation	Clinical sign	HP:0002960 (Autoimmunity); HP:0012647 (Abnormal inflammatory response)	Childhood	Core of "autoinflammation" label

Severity/progression. Variable but potentially severe (combined immunodeficiency plus organ inflammation); course is chronic with episodic inflammatory flares. **Quality of life** is substantially affected by recurrent infections and chronic pneumonitis/colitis, though formal QoL instrument data (EQ-5D/SF-36) are not available for this rare disease.

4. Genetic / Molecular Information

Causal gene. *PIK3CG* (HGNC:8978; OMIM 601232; UniProt P48736), 7q22.3. Encodes p110 γ , the ~1102-residue class IB PI3K catalytic subunit, which partners with regulatory subunits p101 (*PIK3R5*) and p84/p87 (*PIK3R6*).

Pathogenic variants (F008). Missense (p.Arg49Ser, p.Asn1085Ser, p.Arg1021Pro), nonsense (p.Arg849Ter, p.Lys71Ter), and frameshift/complex in-cis deletion (c.[391_397del;456_468del]) alleles are curated to IMD97 in ClinVar (transcript NM_001282426.2). Classifications range from Pathogenic to Likely Pathogenic (one Conflicting).

Variant class / consequence. All are loss-of-function or catalytically inactivating. The p.Arg1021Pro missense allele exemplifies inactivation via disruption of C-terminal regulatory dynamics (F004).

Allele frequency / constraint. gnomAD: pLI $\approx$ 0, LOEUF $\approx$ 0.67, missense Z = 1.53 (F007) — heterozygous LOF is tolerated, consistent with recessive inheritance and healthy carriers. Individual pathogenic alleles are rare/private.

Somatic vs germline. Germline (F001, "germline biallelic *PIK3CG* mutations," PMID: 33054089). (Note: somatic *PIK3CG* alterations arise in cancer contexts but are unrelated to IMD97.)

Modifier genes / epigenetics / chromosomal abnormalities. None specifically established for IMD97 (data not available). Microbiota acts as the principal non-genetic modifier (Section 2).

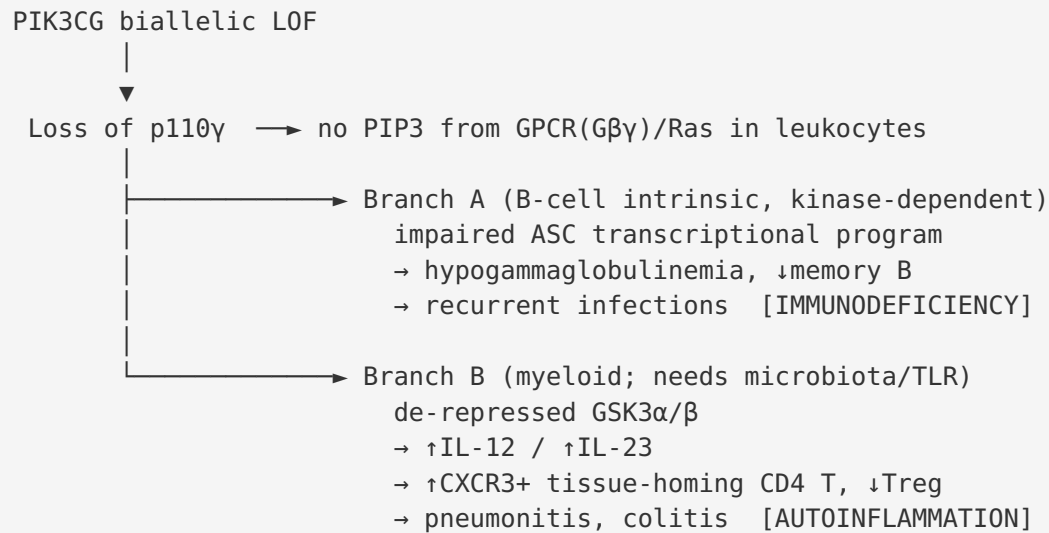
5. Environmental Information

- **Environmental/toxin factors:** None specifically implicated beyond microbial exposure.
- **Lifestyle factors:** Not established.
- **Infectious agents:** Two distinct roles. (1) *Pathogens as opportunists* — recurrent bacterial sinopulmonary infections result from the antibody defect. (2) *Commensal microbiota as an autoinflammation trigger* — TLR-sensed microbial products drive GSK3-dependent IL-12/IL-23 overproduction; the disease is microbiota-dependent in the mouse model ([PMID: 31554793](#)). CHEBI-relevant mediators: lipopolysaccharide (TLR ligand); PIP3 (phosphatidylinositol-3,4,5-trisphosphate, CHEBI:16618 class).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic LOF variants in *PIK3CG*** (missense inactivating, nonsense, or frameshift) **lead to** absence or catalytic inactivation of the p110 γ subunit of PI3Ky (F001, F004; demonstrated).
2. Loss of p110 γ **results in** failure to generate PIP3 downstream of GPCR (G β γ) and Ras stimulation in leukocytes (F005, F006; demonstrated biochemically).
3. **Branch A — Humoral immunodeficiency:** Loss of kinase-dependent PI3Ky signaling *within activated B cells* **results in** failure of the transcriptional program driving antibody-secreting-cell differentiation (F003; demonstrated in human cells/organoids) → **leads to** hypogammaglobulinemia, poor specific/vaccine antibody responses, reduced memory B cells → **leads to** recurrent sinopulmonary infections (clinical).
4. **Branch B — Myeloid autoinflammation:** Loss of p110 γ in macrophages/monocytes **de-represses GSK3 α/β** so that TLR stimulation (triggered by microbiota) **results in** excessive IL-12 and IL-23 production (F002; demonstrated) → **leads to** skewed inflammatory adaptive responses (increased CXCR3⁺ tissue-homing CD4 T cells; reduced Tregs) → **leads to** tissue-infiltrative immunopathology: T-lymphocytic pneumonitis and colitis (clinical).
5. **Environmental gate:** Branch B **requires** exposure to natural microbiota — in mice, disease manifests only after microbial exposure (F002; demonstrated). This is the G $\times$ E node.
6. **Integration:** The simultaneous loss of a positive B-cell differentiation signal (Branch A) and loss of an inflammation-restraining function in myeloid cells (Branch B) **produces** the paradoxical combined immunodeficiency-plus-autoinflammation phenotype (F005; demonstrated/inferred).



Molecular pathways: class I PI3K–AKT/PIP3 signaling (KEGG hsa04151); GPCR signaling; GSK3-dependent cytokine regulation; TLR signaling; IL-12/IL-23–Th1/Th17 axis (Reactome PI3K/AKT; WikiPathways PI3K-AKT).

Cellular processes (GO suggestions): GO:0006954 (inflammatory response); GO:0030593 (neutrophil chemotaxis); GO:0002250 (adaptive immune response); GO:0002377 (immunoglobulin production); GO:0042113 (B cell activation); GO:0016477 (cell migration); GO:0043491 (PI3K/PKB signal transduction).

Cell types (CL suggestions): CL:0000235 (macrophage); CL:0000576 (monocyte); CL:0000775 (neutrophil); CL:0000236 (B cell); CL:0000786 (plasma cell/ASC); CL:0000815 (regulatory T cell); CL:0000624 (CD4⁺ T cell); CL:0000625 (CD8⁺ T cell).

Protein dysfunction: Loss/inactivation of p110γ catalytic (kinase) activity; C-terminal (R1021) destabilization abolishes catalysis (F004).

Immune system involvement: Combined — humoral immunodeficiency + autoinflammation/immune dysregulation; loss of Treg-mediated tolerance contributes.

Molecular profiling: Human data are limited to immunophenotyping (reduced memory B, memory CD8⁺ T, Treg; increased CXCR3⁺ CD4 T) and ex vivo cytokine assays (↑IL-12/IL-23). No large-scale transcriptomic/proteomic/metabolomic patient datasets are available (gap). Lanahan et al. used tonsillar organoids and B-cell transcriptional readouts to define the ASC program dependence (PMID: 38961274).

7. Anatomical Structures Affected

- **Organ/system level:** Immune/hematopoietic system (primary); respiratory system — lungs (pneumonitis, UBERON:0002048); digestive system — colon/intestine (colitis, UBERON:0001155 colon); lymphoid organs (spleen, lymph nodes, tonsil UBERON:0002372); bone marrow/blood (cytopenias, UBERON:0002371).
- **Tissue/cell level:** Mucosal epithelium and lamina propria of gut and airway; leukocyte populations — macrophages, monocytes, neutrophils, B cells/ASCs, CD4/CD8 T cells, Tregs (CL terms in Section 6).
- **Subcellular (GO cellular component):** Plasma membrane / inner leaflet (site of PIP3 generation, GO:0005886); cytosol (GO:0005829); p110 γ is a cytoplasmic/membrane-recruited kinase.
- **Localization / lateralization:** Systemic immune involvement; organ inflammation (lung, colon) is typically **bilateral/diffuse** rather than focal.

8. Temporal Development

- **Onset:** Childhood-onset in reported patients (F001) — pediatric, insidious to subacute, with chronic infections and inflammatory disease.
- **Progression:** Chronic and lifelong; the autoinflammatory component is episodic/fluctuating and microbiota-driven, while the antibody deficiency is persistent.
- **Course pattern:** Chronic with relapsing inflammatory flares superimposed on a stable underlying immunodeficiency.
- **Critical periods / intervention windows:** Early diagnosis enables immunoglobulin replacement and infection prophylaxis; controlling microbial/inflammatory triggers may attenuate tissue disease (inferred).

9. Inheritance and Population

- **Inheritance:** Autosomal recessive (F001, F007). Biallelic LOF required; heterozygous carriers healthy (pLI $\approx$ 0).
- **Penetrance/expressivity:** Presumed high penetrance for biallelic complete LOF, with variable expressivity across the immunodeficiency–autoinflammation spectrum (small n limits precision).
- **Epidemiology:** Ultra-rare; only a handful of families reported worldwide since 2019. Formal prevalence/incidence figures are **not available** (too few cases). No dedicated Orphanet prevalence class identified.

- **Consanguinity/founder effects:** Consanguinity increases likelihood of homozygous LOF (general recessive IEI principle); no specific founder allele established.
- **Carrier frequency:** Not formally estimated; individual pathogenic alleles are rare/private in gnomAD.
- **Demographics:** No established ethnic predisposition or sex bias (the index case was female; too few patients to define a sex ratio).

10. Diagnostics

- **Laboratory tests:** Serum immunoglobulins (IgG/IgA/IgM — hypogammaglobulinemia); specific antibody titers/vaccine responses (impaired); complete blood count (cytopenias); lymphocyte immunophenotyping (reduced memory B, memory CD8⁺ T, Treg; increased CXCR3⁺ CD4 T). LOINC-mappable panels: quantitative immunoglobulins, lymphocyte subsets.
- **Functional immunology:** Ex vivo TLR-stimulated monocyte/macrophage cytokine assays may show elevated IL-12/IL-23 (research-grade; F002). Assessment of PIP3 generation / kinase activity is research-level.
- **Imaging:** Chest CT for interstitial/lymphocytic pneumonitis; GI imaging/endoscopy with biopsy for colitis.
- **Biopsy/pathology:** Lung and colon biopsies show lymphocytic infiltration/inflammation (F001).
- **Genetic testing (definitive):** WES or WGS, or targeted IEI/primary-immunodeficiency gene panels including *PIK3CG*, with confirmation of biallelic LOF variants; single-gene *PIK3CG* sequencing when the phenotype is recognized. ClinVar-curated alleles (F008) guide classification (ACMG/AMP). Segregation analysis confirms compound-het/homozygous state.
- **Clinical criteria / classification:** Recognized as a distinct IEI by the IUIS Expert Committee (2022 and subsequent updates; [PMID: 35748970](#), [PMID: 41608114](#)).
- **Differential diagnosis:** Common variable immunodeficiency (CVID) and other primary antibody deficiencies ([PMID: 34153571](#)); combined immunodeficiencies with immune dysregulation; **APDS (PIK3CD/PIK3R1 — gain-of-function, distinct mechanism and therapy)**; other monogenic IBD/autoinflammatory syndromes. Genetic testing distinguishes IMD97 definitively.
- **Screening:** Cascade genetic testing of relatives for the familial *PIK3CG* alleles; carrier testing for reproductive counseling. No newborn-screening program targets IMD97.

11. Outcome / Prognosis

Formal survival/mortality statistics are **not available** given the tiny cohort. Prognosis depends on control of infections (via immunoglobulin replacement and prophylaxis) and management of organ-specific inflammation (pneumonitis, colitis). Complications include chronic lung disease, chronic/refractory colitis, cytopenia-related morbidity, and infection-related events. With early diagnosis and standard IEI supportive care, meaningful morbidity reduction is expected, but recovery potential and long-term outcomes are uncharacterized (gap). Prognostic biomarkers are not established; immunoglobulin levels, infection frequency, and inflammatory activity serve as pragmatic monitoring measures.

12. Treatment

No IMD97-specific approved therapy or clinical trial exists (ultra-rare disease). Management follows general IEI principles and mechanistic rationale:

Modality	Rationale / target	NCIT suggestion
Immunoglobulin replacement (IVIG/SCIG)	Correct antibody deficiency, reduce infections	NCIT:C561 (Immunoglobulin Therapy)
Antimicrobial prophylaxis	Prevent recurrent sinopulmonary infection	NCIT:C15839 (Antibiotic Therapy)
Immunosuppressive/anti-inflammatory agents (e.g., corticosteroids)	Control pneumonitis/colitis autoinflammation	NCIT:C15267 (Immunosuppressive Therapy)
Anti-cytokine biologics (anti-IL-12/IL-23, e.g., ustekinumab) — <i>mechanistically rational, not validated in IMD97</i>	Neutralize the elevated IL-12/IL-23 axis (F002)	NCIT:C2140 (Monoclonal Antibody Therapy)
Hematopoietic stem cell transplantation (HSCT) — <i>conceptually curative for a hematopoietic-intrinsic IEI; not established for IMD97</i>	Replace defective leukocyte compartment	NCIT:C15431 (Hematopoietic Stem Cell Transplantation)

Pharmacogenomics / gene therapy: No genotype-specific pharmacogenomic data. Gene therapy is theoretical only. Note that because *p110 γ* has a *positive* B-cell-intrinsic role (F003) yet a *restraining* role in myeloid inflammation (F002/F005), pharmacologic PI3K γ **inhibitors** (developed for cancer/inflammation) are **contraindicated** in this LOF disease — they would worsen it. This asymmetry is a key precision-medicine caveat.

13. Prevention

- **Primary prevention:** Not possible for the genetic disease; **genetic counseling** for at-risk families (recessive recurrence risk 25% per pregnancy for carrier couples), carrier testing, and reproductive options (prenatal/preimplantation genetic testing).
- **Secondary prevention:** Early molecular diagnosis via IEI gene panels/WES enables prompt immunoglobulin replacement and prophylaxis before end-organ damage.
- **Tertiary prevention:** Infection prophylaxis, vaccination-adjusted strategies (recognizing impaired antibody responses), and anti-inflammatory management to prevent progressive lung/GI damage.
- **Counseling:** Cascade testing of relatives; genetic counseling per NSGC/ACMG frameworks.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** Mouse *Pik3cg* (NCBI Gene 30955) is the principal ortholog used to model the disease (NCBI Taxon 10090). Human *PIK3CG* NCBI Gene 5294 (Taxon 9606).
- **Natural disease:** No well-characterized naturally occurring *PIK3CG*-deficiency disease in companion animals or wildlife is documented (data not available); the disease is understood chiefly through engineered mouse knockouts.
- **Comparative biology:** PI3K γ function is evolutionarily conserved; mouse and human share the GPCR-activated, leukocyte-enriched biology, though the human disease required the microbiota-humanized (pet-store co-housing) model to fully recapitulate (F002).
- **Zoonotic potential:** Not applicable (non-infectious Mendelian disorder).

15. Model Organisms

- **Primary model:** *Pik3cg* knockout mouse (constitutive and kinase-dead alleles). This model recapitulates myeloid trafficking defects, colitis susceptibility, altered mucosal IgA/IL-10/occludin and microbiota, neutrophil NET/pyroptosis defects, and Treg/Th17 imbalance (PMID: 41794063, PMID: 36031460, PMID: 39024551, PMID: 36243853).

- **Key model characteristic — microbiota dependence:** *Pik3cg*-deficient mice recapitulate major features of human disease **only after exposure to natural microbiota** (co-housing with pet-store mice) (PMID: 31554793) — a critical insight into the environmental requirement and a limitation of clean SPF-housed models.
- **In vitro / cellular models:** Human tonsillar organoids and primary B-cell systems demonstrated the cell-intrinsic ASC-differentiation defect (PMID: 38961274); PI3K γ -deficient HT-29/B6 human colonic epithelial cells modeled barrier/TNF α effects (PMID: 35725890); recombinant p110 γ biophysics defined R1021 inactivation (PMID: 33661099).
- **Resources:** MGI (mouse *Pik3cg*); IMPC/KOMP knockout lines.
- **Limitations:** SPF-housed knockouts under-represent the autoinflammatory phenotype; mouse–human differences in B-cell PI3K γ dependence require human-cell validation.

Mechanistic Model / Interpretation

IMD97 is best understood as the **loss of a single leukocyte-enriched signaling node (p110 γ /PIP3) that simultaneously performs two opposite immunological jobs**: it *drives* a beneficial adaptive output (B-cell $\rightarrow$ antibody-secreting-cell differentiation) and *restrains* a harmful innate output (myeloid IL-12/IL-23 via GSK3). Removing p110 γ therefore subtracts a needed "go" signal from B cells and subtracts a needed "brake" from macrophages at the same time. The result is the counterintuitive coexistence of immunodeficiency and autoinflammation in one patient.

The two arms differ in their environmental gating. **Branch A (humoral)** is cell-intrinsic and largely environment-independent — it manifests as constitutive antibody failure. **Branch B (autoinflammatory)** is conditional: it needs a microbial/TLR trigger to unmask the GSK3-dependent IL-12/IL-23 excess, which is why the mouse model only "worked" after microbiota exposure. This G $\times$ E structure explains why tissue disease is episodic and localized to microbe-rich barrier organs (lung, gut).

Feature	Branch A: Humoral immunodeficiency	Branch B: Myeloid autoinflammation
Cell type	B cells / ASCs	Macrophages, monocytes
p110 γ role lost	Positive ASC-differentiation signal (kinase-dependent)	Restraint of GSK3-driven IL-12/IL-23
Environmental gate	Constitutive (intrinsic)	Requires microbiota/TLR trigger
Clinical output	Hypogammaglobulinemia, infections	Pneumonitis, colitis
Key reference	PMID: 38961274	PMID: 31554793

A crucial translational corollary: because the phenotype is **loss-of-function**, therapeutic PI3K γ **inhibitors** (an active oncology/inflammation drug class) would aggravate IMD97 — the opposite of the APDS situation, where PI3K δ *inhibitors* treat a gain-of-function disease. Rational therapy for IMD97 instead targets *downstream consequences*: immunoglobulin replacement for the antibody defect, and IL-12/IL-23 blockade or general anti-inflammatory therapy for the autoinflammation.

Evidence Base

PMID	Title (abbrev.)	Contribution	Evidence type
31554793	Human PI3Ky deficiency and microbiota-dependent mouse model	Index case; GSK3-dependent IL-12/IL-23; microbiota-dependent model (F001, F002, F005)	Human + mouse
33054089	Germline biallelic <i>PIK3CG</i> mutations, multifaceted immunodeficiency	Independent confirmation; germline AR (F001)	Human
38961274	PI3Ky in B cells promotes antibody responses	Cell-intrinsic ASC-differentiation defect (F003)	Human + organoid
33661099	Disease mutations disrupt PI3Ky C-terminal dynamics	R1021 inactivation mechanism (F004)	Biophysical
41794063	Hematopoietic PI3Ky / macrophage trafficking	Myeloid-predominant expression & trafficking role (F005)	Mouse
36031460	PI3Ky-KO colitis / mucosa / microbiota	Barrier & microbiota phenotypes (F005)	Mouse
39024551	PI3Ky, NETs, noncanonical pyroptosis	Neutrophil effector role (F005)	Mouse
40925445	Class I PI3K regulation review	Framework: PIP3, immune function (F006)	Review
40595568	PI3Ky controls CD8 T-cell trafficking	GPCR-driven T-cell trafficking (F006)	In vitro/mouse
42148525	Ras-PI3Ky interface	Ras activation of p110γ (F006)	Structural
35748970 / 41608114	IUIS IEI classification updates	Formal recognition as distinct IEI	Consensus classification
34153571	Monogenic causes of antibody deficiency	Differential diagnosis context (CVID vs monogenic PAD)	Review

Supporting mouse studies across diverse disease contexts (abdominal aortic aneurysm, kidney injury, periodontitis, dengue, Zika, psoriasis) consistently show PI3Ky as a pro-inflammatory, myeloid-trafficking effector — reinforcing that its **loss** removes both inflammatory drive and inflammatory restraint depending on cell context.

Limitations and Knowledge Gaps

1. **Extremely small human cohort.** Disease definition rests on a handful of index patients; frequencies, penetrance, expressivity, epidemiology (prevalence/incidence), and prognosis statistics cannot be quantified.
 2. **No formal Orphanet/ICD codes** identified for IMD97 as a standalone entity; classification currently sits under broader IEI umbrellas.
 3. **No treatment trials.** All therapeutic recommendations are mechanistic/extrapolated; response rates and adverse-event data specific to IMD97 are unavailable.
 4. **Molecular profiling gap.** No large-scale patient transcriptomic/proteomic/metabolomic datasets exist; mechanism relies on immunophenotyping, ex vivo cytokine assays, organoids, and mouse genetics.
 5. **Model translation.** SPF mouse knockouts under-represent autoinflammation; microbiota-humanized models are needed to fully mirror human disease. Mouse–human differences in B-cell PI3Ky dependence require ongoing human-cell validation.
 6. **Variant interpretation.** Some ClinVar alleles are "Conflicting" or "Likely Pathogenic"; functional validation (as done for R1021) is not available for all variants.
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Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** (e.g., via IUIS/ESID networks) to aggregate cases, define phenotype frequencies, natural history, and treatment responses.
2. **Systematic functional characterization** of all ClinVar *PIK3CG* alleles (kinase activity, PIP3 generation, protein stability) to firm up ACMG classifications — extending the R1021 HDX-MS approach ([PMID: 33661099](https://pubmed.ncbi.nlm.nih.gov/33661099/)).

3. **Therapeutic proof-of-concept for IL-12/IL-23 blockade** (e.g., ustekinumab) in IMD97 patients or in microbiota-humanized *Pik3cg*-KO mice, testing whether targeting the GSK3-driven cytokine axis controls pneumonitis/colitis.
 4. **Single-cell multi-omics of patient B cells and myeloid cells** to map the divergent transcriptional consequences of p110 γ loss (validate the ASC-program defect and the myeloid IL-12/IL-23 de-repression in humans).
 5. **Evaluate HSCT outcomes** as a potentially curative option for severe combined immunodeficiency-plus-autoinflammation presentations.
 6. **Explicitly document contraindication of PI3Ky inhibitors** in clinical guidance to prevent iatrogenic worsening, given the LOF nature of the disease.
 7. **Formal ontology/coding submissions** to assign dedicated Orphanet and ICD-11 identifiers and complete HPO annotation with frequencies as cohort data accrue.
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Report compiled from 5 investigation iterations, 8 confirmed findings, and 54 reviewed papers. Evidence types are labeled throughout: human clinical, model organism, in vitro/organoid, biophysical/structural, and population genomics.
